

# Story of Arithmetic

Part 1





## About the Course

This course gives students a glimpse into how the ideas that guide our study of arithmetic were first discovered.



## Placement & Combining Tips

This course is intended as a precursor to Part 2; however, the order may be reversed to facilitate combining students.



## Scheduling

| GRADE | SCHEDULE INFO.                                    | BOOKS                                                                                                                                          |
|-------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | Story of Arithmetic: Part 1<br>1 time/week 15 min | Number Stories of Long Ago<br>The Student Bible Atlas, Revised Edition<br>Historical Atlas of the World 6th Ed.<br>Student World Atlas 6th Ed. |

### Sample Weekly View

| Day 1                              | Day 2 | Day 3 | Day 4 | Day 5 |
|------------------------------------|-------|-------|-------|-------|
| <b>Story of Arithmetic: Part 1</b> |       |       |       |       |
| Story of Arithmetic: Part 1        |       |       |       |       |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Story of Arithmetic: Part 1



Number Stories of Long Ago



The Student Bible Atlas, Revised Edition



Historical Atlas of the World 6th Ed.



Student World Atlas 6th Ed.



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

(No Quick Links Assigned)

Click THIS text  
or scan the QR  
code for links.



# Story of Arithmetic: Part 1

| How To Teach                                                                        |                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Recap</b><br>Connect back to previous lesson.                                                                                                                                                                    |
|    | <b>Introduce</b><br>Optional: Use a picture, map, or engaging question to prepare the student to receive the idea from the day's lesson. Talk about one or two unknown words that will be crucial to understanding. |
|    | <b>Read</b><br>Read the new passage out loud or have the student read it orally or silently.                                                                                                                        |
|    | <b>Narrate</b><br>Have the student narrate orally or in writing.                                                                                                                                                    |
|   | <b>Discuss</b> <ul style="list-style-type: none"><li>• Give students an opportunity to share their thoughts on the lesson and any connections made.</li></ul>                                                       |
|  | <b>Practice</b><br>Optional: Attempt to recreate some of the problems from the stories.                                                                                                                             |

# Story of Arithmetic: Part 1

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## Term 1

### WEEK 1 15m Story of Arithmetic: Part 1- Lesson 1

*Counting: Introduction*

Materials: Number Stories of Long Ago, Student World Atlas

#### → INTRO

Today, we will meet a boy who lived long ago at the foot of "Mt. Yu," or Mount Yushan, which is in Taiwan. Find Taiwan on the map. What kind of climate does it have?

Map Resource: Student World Atlas "Asia" p.92-95

#### → READ, NARRATE, & DISCUSS

p.7-8 "The logs are" - "to do so."

### WEEK 2 15m Story of Arithmetic: Part 1- Lesson 2

*How Ching, An-Am + Menes Counted*

Materials: Number Stories of Long Ago, Historical Atlas of the World, World Wall Map

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.8-10 "At the time" - "great many apples."

#### → LOCATE

Map Resource: Historical Atlas of the World "Alexander's Empire: 336-323 B.C." p.10-11

Find ancient Mesopotamia and Babylon.

Map Resource: World Wall Map

What is this area (ancient Mesopotamia and Babylon) called today? Now find Egypt. Who lived closer to An-am, Menes or Ching?

### WEEK 3 15m Story of Arithmetic: Part 1- Lesson 3

*Counting Larger Numbers*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.10-11 "When Ching and" - "were quite enough."

#### → PRACTICE

If the boys could only count to "three threes and two," what is the maximum number of sheep they could count?

### WEEK 4 15m Story of Arithmetic: Part 1- Lesson 4

*Counting Even Larger Numbers*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.11-12 "While the boys" - "count by tens."

# Story of Arithmetic: Part 1

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## Term 1

### WEEK 5 15m Story of Arithmetic: Part 1- Lesson 5

*Representing Numbers by the Hand*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.12-13 "Near the equator" - "our two hands."

### WEEK 6 15m Story of Arithmetic: Part 1- Lesson 6

*Representing Numbers (cont.)*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.13-14 "Long after the" - "the Story-Teller."

#### → VIEW & DISCUSS

The picture on p.10 shows how people represented numbers using their hands. Watch this video showing the numbers 1-30 in American Sign Language. Can you follow the patterns?

∞ Video Link: Numbers 1-30 ASL (1:45)

### WEEK 7 15m Story of Arithmetic: Part 1- Lesson 7

*Writing Numbers: Introduction*

Materials: Number Stories of Long Ago, Student World Atlas

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.17-18 "Just one little" - "as you do."

#### → LOCATE

Map Resource: Student World Atlas "Asia" p.92

Find the Hwang Ho River; the atlas calls it the Yellow (Huang) River. Use the scale to estimate how far Chang's home was from Ching's.

### WEEK 8 15m Story of Arithmetic: Part 1- Lesson 8

*Numbers in Other Languages*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.18-21 "Then his father" - "going to school."

#### → STUDY

Look at the number 789 in Chinese and read the caption. Now look at the Chinese figures. Can you see the pattern?

Image Resource: Number Stories of Long Ago "Chinese Number; Chinese Figures" p.19, 18

#### • IMPORTANT DATES

Ancient Babylonian Numerals (c.1800 BC)

# Story of Arithmetic: Part 1

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| Term 1         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>WEEK 9</b>  | <div><div><div>15m Story of Arithmetic: Part 1- Lesson 9</div><div><i>Egyptian Numerals</i></div><div>Materials: Number Stories of Long Ago</div><div>→ <b>RECAP</b><br/>What did you read last time?</div><div>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.21-22 "While Chang" - "invented something better."</div></div></div>                                                                                                                                                                                                                                                                                                                             | <div><div>• <b>IMPORTANT DATES</b><br/>Papyrus Invented (c.3000 BC)<br/>Temple at Luxor (1400 BC, Egypt)</div></div> |
| <b>WEEK 10</b> | <div><div><div>15m Story of Arithmetic: Part 1- Lesson 10</div><div><i>Learning More About Arithmetic</i></div><div>Materials: Number Stories of Long Ago</div><div>→ <b>RECAP</b><br/>What did you read last time?</div><div>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.22-23 "There was a" - "ever be needed."</div><div>→ <b>STUDY</b><br/>Look at the Egyptian number for 27,529. Can you see a pattern? How might you write 37,354? What problem would you encounter if you tried adding these two numbers using the Egyptian symbols? How could you solve it?<br/>Image Resource: Number Stories of Long Ago "Egyptian Number" p.23</div></div></div> |                                                                                                                      |
| <b>WEEK 11</b> | <div><div><div>15m Story of Arithmetic: Part 1- Lesson 11</div><div><i>The Oldest Book on Mathematics</i></div><div>Materials: Number Stories of Long Ago</div><div>→ <b>RECAP</b><br/>What did you read last time?</div><div>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.23-24 "Ahmes lived nearly" - "time for bed."</div></div></div>                                                                                                                                                                                                                                                                                                                     | <div><div>• <b>IMPORTANT DATES</b><br/>Oldest Book on Mathematics<br/>Written by Ahmes (1700 BC, Egypt)</div></div>  |
| <b>WEEK 12</b> | <div><div><div>15m Story of Arithmetic: Part 1- Lesson 12</div><div><i>Exam Week</i></div><div>Materials: Number Stories of Long Ago</div><div>→ <b>EXAMS</b><br/>Answer question(s) related to course.</div><div>Questions will come from:<br/>Number Stories of Long Ago</div></div></div>                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                      |

# Story of Arithmetic: Part 1

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## Term 2

### WEEK 1 15m Story of Arithmetic: Part 1- Lesson 13

*Ancient Coins*

Materials: Number Stories of Long Ago, Historical Atlas of the World, World Wall Map or Student World Atlas

#### → RECAP

What do you remember from last term?

#### → READ, NARRATE, & DISCUSS

p.27-29 "What is the" - "to use paper."

#### → LOCATE

Map Resource: Historical Atlas of the World "Classical Greece About 450 B.C." p.7

Pergamon was also called "Pergamum." It was located in Mysia. Find it in the atlas.

Map Resource: World Wall Map or Student World Atlas

Find on a current world map where ancient Pergamon was located. What is this place called now?

#### • IMPORTANT DATES

Earliest Money Used (c.550 BC, Asia Minor)

### WEEK 2 15m Story of Arithmetic: Part 1- Lesson 14

*New Ways of Writing Numbers*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.29 "Hippias learned to" - "our word alphabet."

#### → STUDY

Look at the Greek numerals and the number 2977 written in Greek. Can you find the pattern? How would you write 3,684?

Image Resource: Number Stories of Long Ago "Greek Numerals; Number Written in Greek" p.30

### WEEK 3 15m Story of Arithmetic: Part 1- Lesson 15

*Greek Letter-Numerals*

Materials: Number Stories of Long Ago, Student Bible Atlas

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.29-31 "While Hippias was" - "division very hard."

#### → LOCATE

Map Resource: Student Bible Atlas "Map 1: Lands of the Bible" p.4-5 Find Athens and Jerusalem on the map. Use the scale to estimate how many kilometers one would have to sail to travel from one to the other.

#### • IMPORTANT DATES

Early Roman Numerals (c.900 BC, Rome)

### WEEK 4 15m Story of Arithmetic: Part 1- Lesson 16

*Late Roman Numerals*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?



# Story of Arithmetic: Part 1

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## Term 2

### → READ, NARRATE, & DISCUSS

p.31-33 "While Hippias was" - "is taken away?"

### → DISCUSS

Which Roman numeral could become ten more if ten were taken away? Is there more than one answer?

#### WEEK 5 15m Story of Arithmetic: Part 1- Lesson 17

*Roman Numeral Puzzles*

Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.33-34 "Titus then asked" - "Story-Teller."

### → PRACTICE

Try some of the tricks that Titus and Caius played using Roman numerals. Can you figure out how they work?

#### WEEK 6 15m Story of Arithmetic: Part 1- Lesson 18

*Intro: Ancient Hindu Numerals*

Materials: Number Stories of Long Ago, Student World Atlas

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.37-38 "No Story Tonight," - "whatever of America."

### → LOCATE

Map Resource: Student World Atlas "Asia" p.92-93  
Find the Indus River on the Physical Map and then on the Political Map.  
What do you notice? How many kilometers is it from Jerusalem?

#### WEEK 7 15m Story of Arithmetic: Part 1- Lesson 19

*Ancient Hindu Numerals*

Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.38-39 "Of all the" - "you are reading."

#### WEEK 8 15m Story of Arithmetic: Part 1- Lesson 20

*Early Numerals*

Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.39 "When Gupta grew" - "write them today."

# Story of Arithmetic: Part 1

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## Term 2

### WEEK 9 15m Story of Arithmetic: Part 1- Lesson 21

#### Arabic Numerals

Materials: Number Stories of Long Ago, Historical Atlas of the World

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.39-42 "About the time" - "observatory at Bagdad."

#### → LOCATE

Map Resource: Historical Atlas of the World "Islamic World c.1000 AD" p.26-27

Find the Tigris River and Baghdad. In which direction would you need to travel to get to the Indus River? To the Mediterranean Sea?

### WEEK 10 15m Story of Arithmetic: Part 1- Lesson 22

#### Old European Numerals

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.42 "Mohammed the son" - "since that time."

#### → STUDY

Look at the Old European Numerals. What are the similarities and differences between these and the ones we use today?

Image Resource: Number Stories of Long Ago "Old European Numerals" p.42

#### • IMPORTANT DATES

Zero Invented (3 BC, Mesopotamia)  
Hindu-Arabic Numerals Invented (c.650 AD, India)  
Arabic Numbers First Used in Europe (c.1000 AD)

### WEEK 11 15m Story of Arithmetic: Part 1- Lesson 23

#### Changes in Numerals

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.43-45 "In France nearly" - "the Story-Teller."

#### → DISCUSS

Why do you think the forms of letters and numbers have stayed about the same since the printing press was invented?

### WEEK 12 15m Story of Arithmetic: Part 1- Lesson 24

#### Exam Week

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Number Stories of Long Ago

# Story of Arithmetic: Part 1

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## Term 3

### WEEK 1 15m Story of Arithmetic: Part 1 - Lesson 25

*Adding Roman Numerals*

Materials: Number Stories of Long Ago

#### → RECAP

What do you remember from last term?

#### → READ, NARRATE, & DISCUSS

p.49-50 "I don't see" - "with our own."

#### → PRACTICE

Try adding LXII + XVII (62 + 17) using Roman numerals and our modern numerals. Do you agree with the Story-Teller that they are about equally difficult? Try subtracting the same numbers.

### WEEK 2 15m Story of Arithmetic: Part 1 - Lesson 26

*Adding With Pebbles*

Materials: Number Stories of Long Ago, AL Abacus (optional)

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.50-52 "After learning to" - "as we are."

#### → STUDY

Look at the pictures on p.51-52 to see if you can figure out how Caius calculated  $22 + 139$  using pebbles. Calculate  $22 + 139$  on your abacus. How is our way easier?

### WEEK 3 15m Story of Arithmetic: Part 1 - Lesson 27

*Roman Abacus*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.52-53 "Of course Caius" - "moved in grooves."

#### • IMPORTANT DATES

Abacus Invented (c.2400 BC, Babylonia)

### WEEK 4 15m Story of Arithmetic: Part 1 - Lesson 28

*Ancient Chinese and Korean Counters*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.54-55 "While Caius was" - "until quite recently."

### WEEK 5 15m Story of Arithmetic: Part 1 - Lesson 29

*Chinese Abacus*

Materials: Number Stories of Long Ago, AL Abacus (optional)

#### → RECAP

What did you read last time?

# Story of Arithmetic: Part 1

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## Term 3

### → READ, NARRATE, & DISCUSS

p.55-56 "It was more" - "of our cities."

### → VIEW & DISCUSS

Learn how the Chinese abacus works. How is it similar to and different from the AL Abacus? Can you figure out what number is represented in the picture on p.55?

∞ Video Link: Abacus Tutorial (3:28)

## WEEK 6 ☐ 15m Story of Arithmetic: Part 1- Lesson 30

*Japanese Abacus*

☐ Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.57 "About three hundred" - "pencil and paper."

### → VIEW & DISCUSS

Learn how the Japanese abacus works. How is it similar to and different from the Chinese abacus? Can you figure out what number is represented in the picture on p.57?

∞ Video Link: How to Add Using a Soroban (2:43) Time Marker 0:40-3:23

## WEEK 7 ☐ 15m Story of Arithmetic: Part 1- Lesson 31

*Counting Tables*

☐ Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.57-59 "About four hundred" - "score in games."

## WEEK 8 ☐ 15m Story of Arithmetic: Part 1- Lesson 32

*Slide Rule*

☐ Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.59-60 "Robert added numbers" - "right or left."

### → VIEW

Learn how people used slide rules for multiplication before they had calculators.

∞ Video Link: Slide Rules Part 1: Multiplication (3:19)

## WEEK 9 ☐ 15m Story of Arithmetic: Part 1- Lesson 33

*Conclusion*

☐ Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

# Story of Arithmetic: Part 1

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| Term 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.61-62 "This shows how" - "the Story-Teller."</p>                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| <p><b>WEEK 10</b> 15m <b>Story of Arithmetic: Part 1 - Lesson 34</b><br/><i>Math Games</i></p> <div>Materials: AL Abacus</div> <p>→ <b>RECAP</b><br/>What did you read last time?</p> <p>→ <b>PRACTICE</b><br/>Choose a game to play in which you can keep score using an abacus (i.e., tic-tac-toe). Try it with the back side of the AL Abacus first. Then, if you like, you can create your own suan pan or soroban by drawing the frame and using beans on top for counters. The first to win three games is the champion!</p> |  |
| <p><b>WEEK 11</b> 15m <b>Story of Arithmetic: Part 1 - Lesson 35</b><br/><i>Catch-Up Day</i></p> <p>→ <b>CATCH UP</b><br/>Use this time to catch up on lessons in this or another course, or explore Extra Helpings.</p>                                                                                                                                                                                                                                                                                                           |  |
| <p><b>WEEK 12</b> 15m <b>Story of Arithmetic: Part 1 - Lesson 36</b><br/><i>Exam Week</i></p> <p>→ <b>EXAMS</b><br/>Answer question(s) related to course.</p> <p>Questions will come from:<br/>Number Stories of Long Ago</p>                                                                                                                                                                                                                                                                                                      |  |

# Story of Arithmetic: Part 1

## Examination

- Term 1** · Number Stories of Long Ago: Tell what you know about how people in ancient Asia and Mesopotamia counted.
- Term 2** · Number Stories of Long Ago: Tell what you know about Greek and Roman numerals. OR Tell what you know about the development of number writing.
- Term 3** · Number Stories of Long Ago: Tell what you know about the history of the abacus.